In [2]:
```python
import pandas as pd
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv(r"C:\Users\acer\Downloads\matches.csv")

# Display first few rows
df.head()
```

Out[2]:

	id	season	city	date	match_type	player_of_match	venue	team1	
0	335982	2007/08	Bangalore	2008-04-18	League	BB McCullum	M Chinnaswamy Stadium	Royal Challengers Bangalore	
1	335983	2007/08	Chandigarh	2008-04-19	League	MEK Hussey	Punjab Cricket Association Stadium, Mohali	Kings XI Punjab	(
2	335984	2007/08	Delhi	2008-04-19	League	MF Maharoof	Feroz Shah Kotla	Delhi Daredevils	Ra
3	335985	2007/08	Mumbai	2008-04-20	League	MV Boucher	Wankhede Stadium	Mumbai Indians	Cha Ba
4	335986	2007/08	Kolkata	2008-04-20	League	DJ Hussey	Eden Gardens	Kolkata Knight Riders	C

In [3]:
```python
print("Rows and Columns:", df.shape)
```

```
Rows and Columns: (1095, 20)
```

In [4]:
```python
df.head()
```

Out[4]:

	id	season	city	date	match_type	player_of_match	venue	team1	
0	335982	2007/08	Bangalore	2008-04-18	League	BB McCullum	M Chinnaswamy Stadium	Royal Challengers Bangalore	
1	335983	2007/08	Chandigarh	2008-04-19	League	MEK Hussey	Punjab Cricket Association Stadium, Mohali	Kings XI Punjab	(
2	335984	2007/08	Delhi	2008-04-19	League	MF Maharoof	Feroz Shah Kotla	Delhi Daredevils	Ra
3	335985	2007/08	Mumbai	2008-04-20	League	MV Boucher	Wankhede Stadium	Mumbai Indians	Cha Ba
4	335986	2007/08	Kolkata	2008-04-20	League	DJ Hussey	Eden Gardens	Kolkata Knight Riders	C

In [5]:
```python
print(df.columns)
```

```
Index(['id', 'season', 'city', 'date', 'match_type', 'player_of_match',
       'venue', 'team1', 'team2', 'toss_winner', 'toss_decision', 'winner',
       'result', 'result_margin', 'target_runs', 'target_overs', 'super_over',
       'method', 'umpire1', 'umpire2'],
      dtype='object')
```

In [6]:
```python
df.dtypes
```

Out[6]:
```
id                   int64
season              object
city                object
date                object
match_type          object
player_of_match     object
venue               object
team1               object
team2               object
toss_winner         object
toss_decision       object
winner              object
result              object
result_margin      float64
target_runs        float64
target_overs       float64
super_over          object
method              object
umpire1             object
umpire2             object
dtype: object
```

In [7]:
```python
print("Unique IDs:", df['id'].nunique())
print("Total Rows:", len(df))
```

```
Unique IDs: 1095
Total Rows: 1095
```

In [8]:
```python
print(df.isnull().sum())
```

```
id                    0
season                0
city                 51
date                  0
match_type            0
player_of_match       5
venue                 0
team1                 0
team2                 0
toss_winner           0
toss_decision         0
winner                5
result                0
result_margin        19
target_runs           3
target_overs          3
super_over            0
method             1074
umpire1               0
umpire2               0
dtype: int64
```

In [9]:
```python
print("Duplicate Rows:", df.duplicated().sum())
```

```
Duplicate Rows: 0
```

In [10]:

```python
# Remove duplicates
df = df.drop_duplicates()

# Fill missing values

df['winner'] = df['winner'].fillna("No Result")
df['player_of_match'] = df['player_of_match'].fillna("Unknown")
df['method'] = df['method'].fillna("Normal")
df['result_margin'] = df['result_margin'].fillna(0)

# Check again
print(df.isnull().sum())
```

```
id                   0
season               0
city                51
date                 0
match_type           0
player_of_match      0
venue                0
team1                0
team2                0
toss_winner          0
toss_decision        0
winner               0
result               0
result_margin        0
target_runs          3
target_overs         3
super_over           0
method               0
umpire1              0
umpire2              0
dtype: int64
```

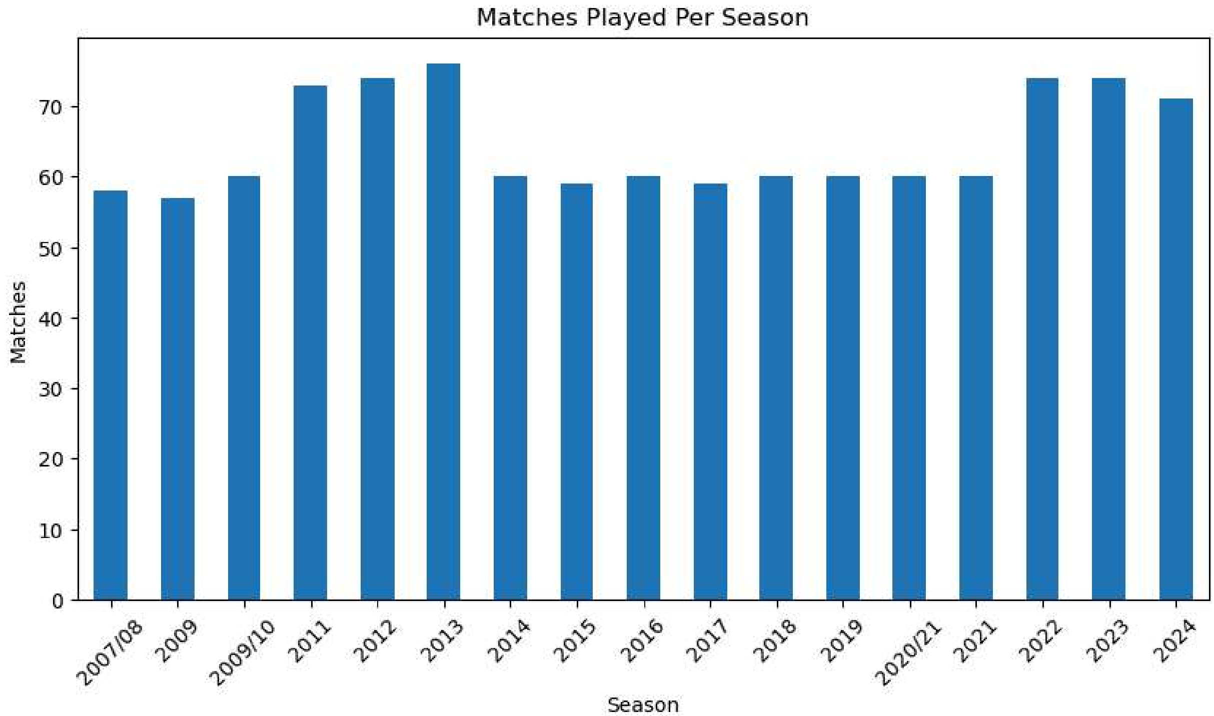
In [11]:
```
matches_per_season = df.groupby('season')['id'].count()

print(matches_per_season)
```

```
season
2007/08    58
2009       57
2009/10    60
2011       73
2012       74
2013       76
2014       60
2015       59
2016       60
2017       59
2018       60
2019       60
2020/21    60
2021       60
2022       74
2023       74
2024       71
Name: id, dtype: int64
```

In [12]:
```
print(matches_per_season.idxmax())
print(matches_per_season.max())
```

```
2013
76
```

In [13]:
```python
plt.figure(figsize=(10,5))
matches_per_season.plot(kind='bar')
plt.title("Matches Played Per Season")
plt.xlabel("Season")
plt.ylabel("Matches")
plt.xticks(rotation=45)
plt.show()
```



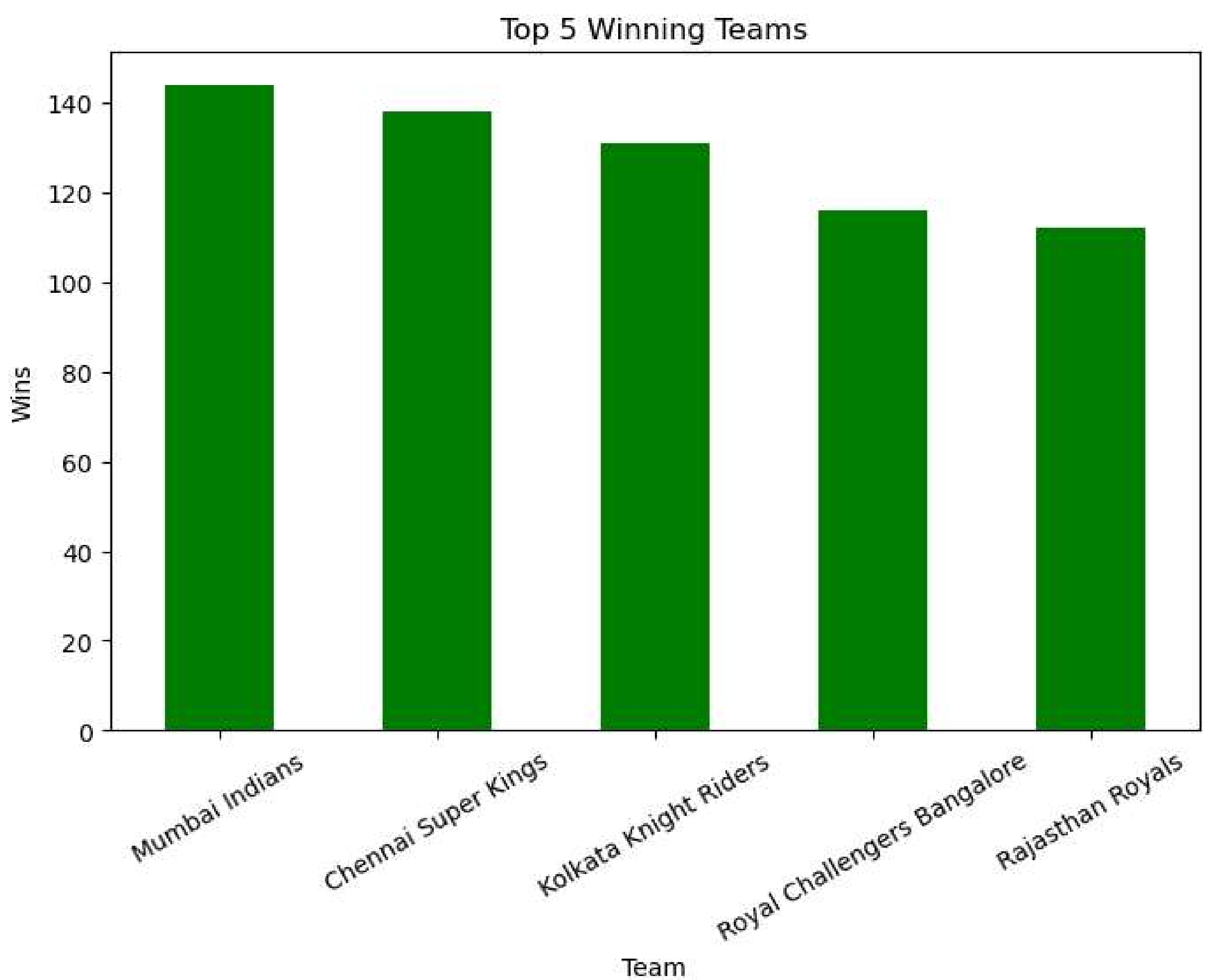
In [14]:
```python
team_wins = df['winner'].value_counts()

print(team_wins)
```

```
Mumbai Indians                 144
Chennai Super Kings            138
Kolkata Knight Riders          131
Royal Challengers Bangalore    116
Rajasthan Royals               112
Kings XI Punjab                 88
Sunrisers Hyderabad             88
Delhi Daredevils                67
Delhi Capitals                  48
Deccan Chargers                 29
Gujarat Titans                  28
Punjab Kings                    24
Lucknow Super Giants            24
Gujarat Lions                   13
Pune Warriors                   12
Rising Pune Supergiant          10
Royal Challengers Bengaluru      7
Kochi Tuskers Kerala             6
Rising Pune Supergiants          5
No Result                        5
Name: winner, dtype: int64
```

In [15]:
```
top5 = team_wins.head(5)

print(top5)
```

```
Mumbai Indians                 144
Chennai Super Kings            138
Kolkata Knight Riders          131
Royal Challengers Bangalore    116
Rajasthan Royals               112
Name: winner, dtype: int64
```

In [16]:
```
plt.figure(figsize=(8,5))
top5.plot(kind='bar', color='green')
plt.title("Top 5 Winning Teams")
plt.xlabel("Team")
plt.ylabel("Wins")
plt.xticks(rotation=30)
plt.show()
```



In [37]:
```python
csk = df[df['winner'] == 'Chennai Super Kings']

csk
```

Out[37]:

	id	season	city	date	match_type	player_of_match	venue	team1
1	335983	2007/08	Chandigarh	2008-04-19	League	MEK Hussey	Punjab Cricket Association Stadium, Mohali	Kings XI Punjab
7	335989	2007/08	Chennai	2008-04-23	League	ML Hayden	MA Chidambaram Stadium, Chepauk	Chennai Super Kings
11	335993	2007/08	Chennai	2008-04-26	League	JDP Oram	MA Chidambaram Stadium, Chepauk	Chennai Super Kings
14	335996	2007/08	Bangalore	2008-04-28	League	MS Dhoni	M Chinnaswamy Stadium	Royal Challengers Bangalore
27	336009	2007/08	Delhi	2008-05-08	League	MS Dhoni	Feroz Shah Kotla	Delhi Daredevils
...	...	...	...	...	...	...	...	...
1045	1426260	2024	Chennai	2024-04-08	League	RA Jadeja	MA Chidambaram Stadium, Chepauk, Chennai	Kolkata Knight Riders
1052	1426267	2024	Mumbai	2024-04-14	League	M Pathirana	Wankhede Stadium, Mumbai	Chennai Super Kings
1069	1426284	2024	Chennai	2024-04-28	League	RD Gaikwad	MA Chidambaram Stadium, Chepauk, Chennai	Chennai Super Kings
1076	1426291	2024	Dharamsala	2024-05-05	League	RA Jadeja	Himachal Pradesh Cricket Association Stadium, ...	Chennai Super Kings
1084	1426299	2024	Chennai	2024-05-12	League	Simarjeet Singh	MA Chidambaram Stadium, Chepauk, Chennai	Rajasthan Royals

138 rows × 22 columns

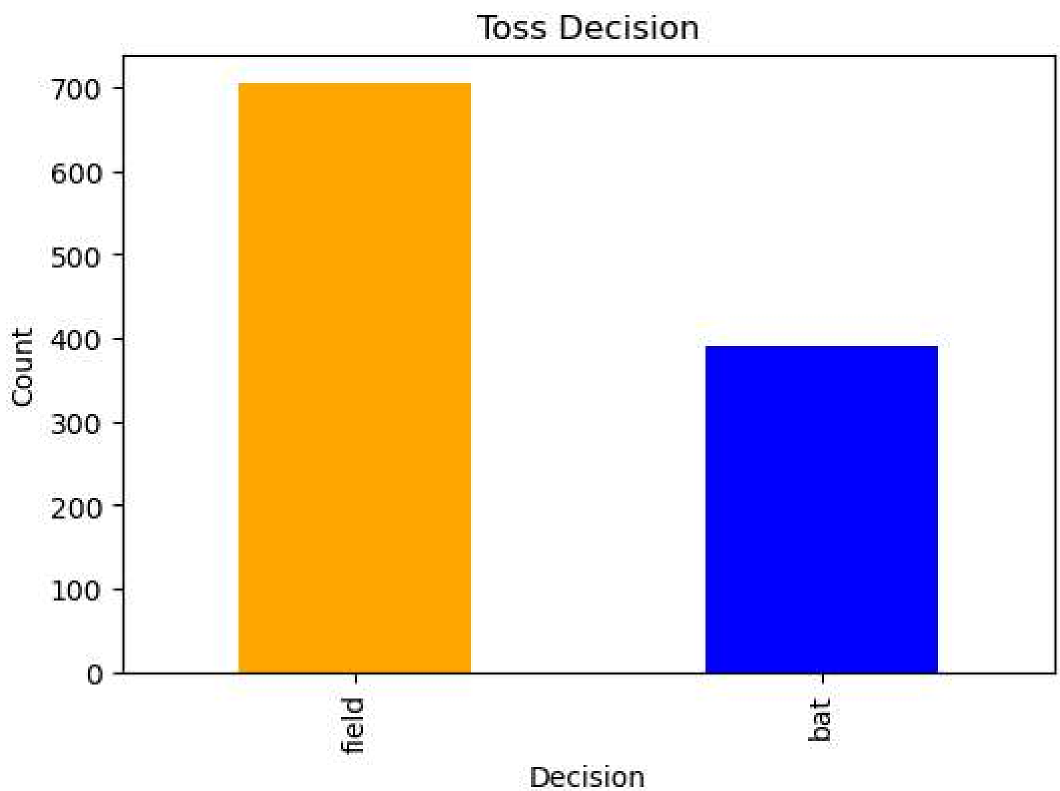
In [18]:
```
print("CSK Wins:", len(csk))
```

```
CSK Wins: 138
```

In [19]:
```
toss = df['toss_decision'].value_counts()

print(toss)
```

```
field    704
bat      391
Name: toss_decision, dtype: int64
```

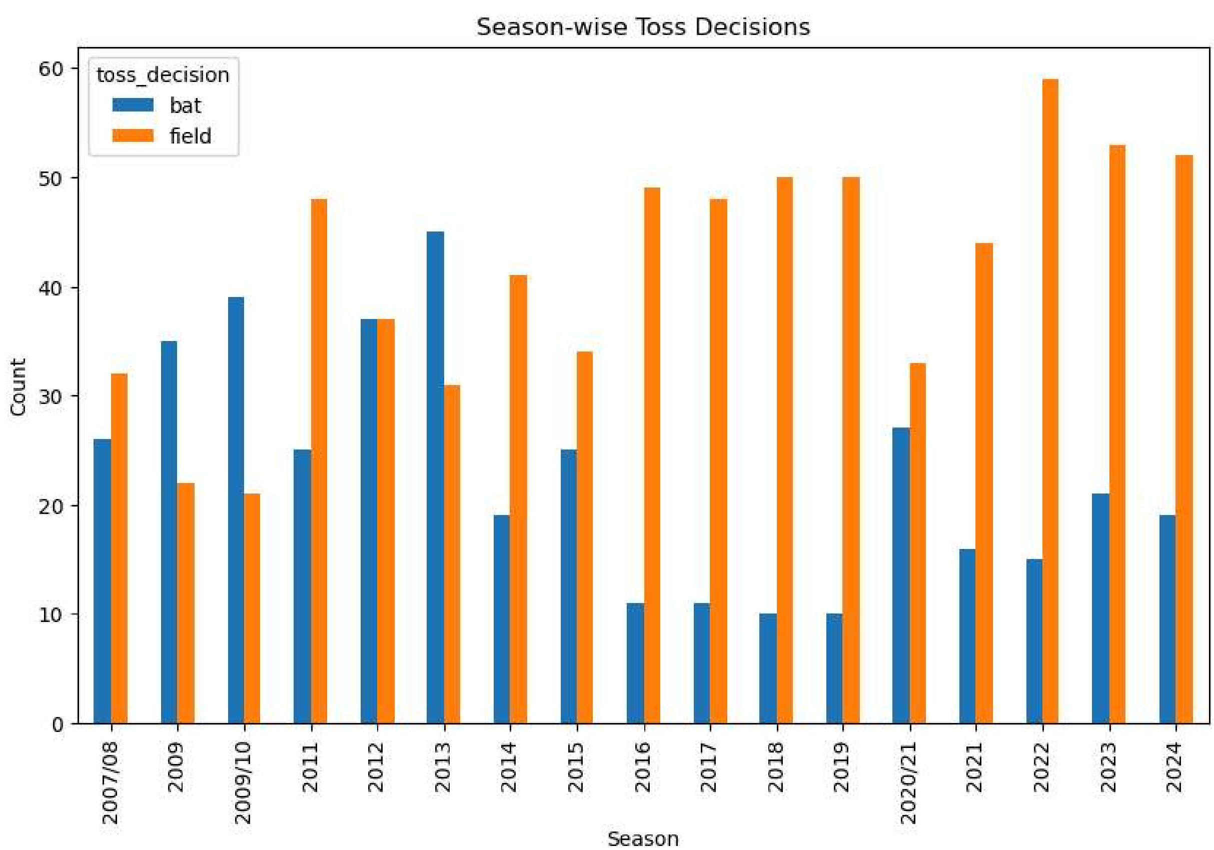
In [20]:
```
plt.figure(figsize=(6,4))
toss.plot(kind='bar', color=['orange','blue'])
plt.title("Toss Decision")
plt.xlabel("Decision")
plt.ylabel("Count")
plt.show()
```



In [21]:
```
pivot = pd.crosstab(df['season'], df['toss_decision'])

print(pivot)
```

```
toss_decision  bat  field
season
2007/08         26     32
2009            35     22
2009/10         39     21
2011            25     48
2012            37     37
2013            45     31
2014            19     41
2015            25     34
2016            11     49
2017            11     48
2018            10     50
2019            10     50
2020/21         27     33
2021            16     44
2022            15     59
2023            21     53
2024            19     52
```

In [22]:
```
pivot.plot(kind='bar', figsize=(10,6))
plt.title("Season-wise Toss Decisions")
plt.xlabel("Season")
plt.ylabel("Count")
plt.show()
```



In [23]:
```
venue = df['venue'].value_counts()

print(venue)
```

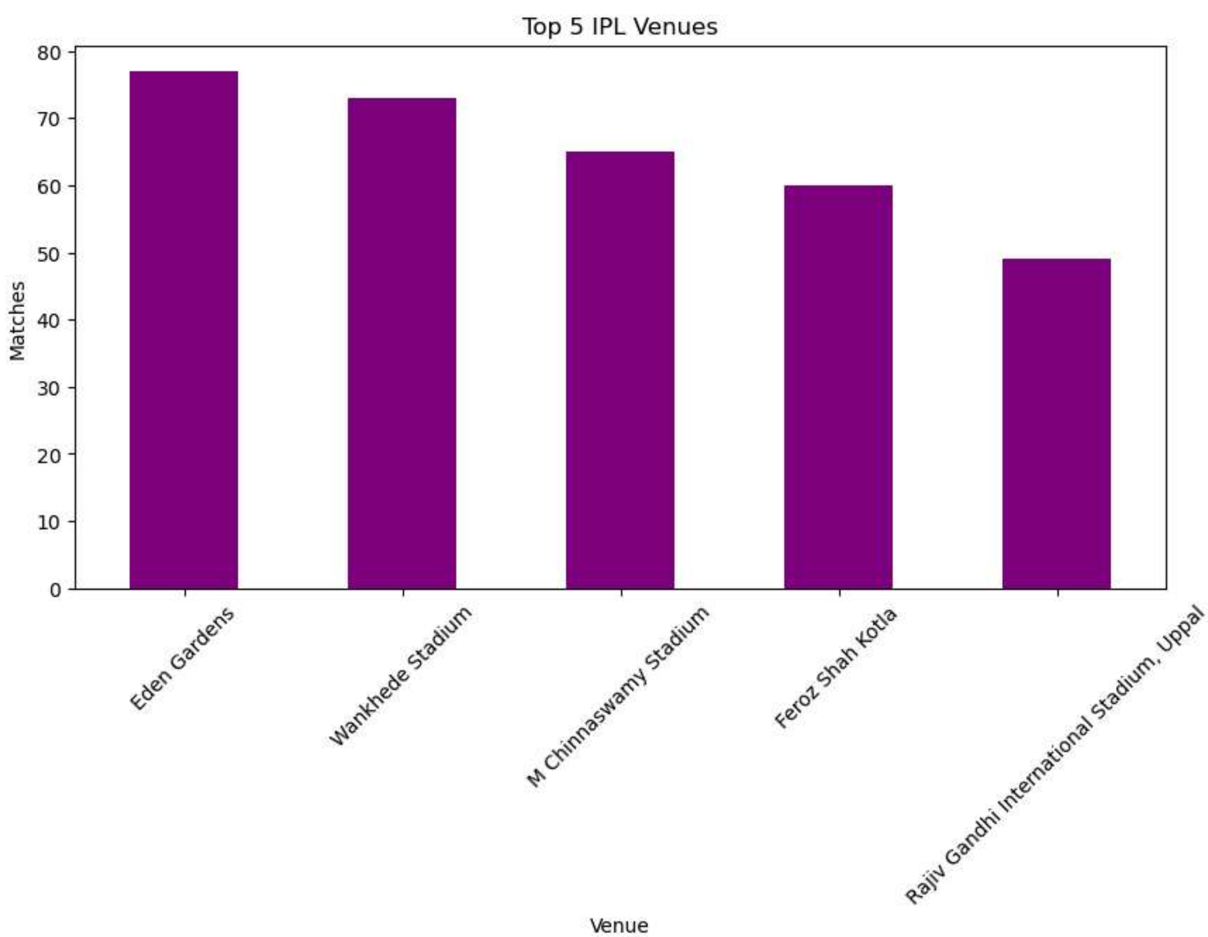
Eden Gardens	77
Wankhede Stadium	73
M Chinnaswamy Stadium	65
Feroz Shah Kotla	60
Rajiv Gandhi International Stadium, Uppal	49
MA Chidambaram Stadium, Chepauk	48
Sawai Mansingh Stadium	47
Dubai International Cricket Stadium	46
Wankhede Stadium, Mumbai	45
Punjab Cricket Association Stadium, Mohali	35
Sheikh Zayed Stadium	29
Sharjah Cricket Stadium	28
MA Chidambaram Stadium, Chepauk, Chennai	28
Narendra Modi Stadium, Ahmedabad	24
Maharashtra Cricket Association Stadium	22
Dr DY Patil Sports Academy, Mumbai	20
Brabourne Stadium, Mumbai	17
Dr DY Patil Sports Academy	17
Eden Gardens, Kolkata	16
Subrata Roy Sahara Stadium	16
Arun Jaitley Stadium, Delhi	16
Rajiv Gandhi International Stadium	15
M.Chinnaswamy Stadium	15
Kingsmead	15
Arun Jaitley Stadium	14
Bharat Ratna Shri Atal Bihari Vajpayee Ekana Cricket Stadium, Lucknow	14
M Chinnaswamy Stadium, Bengaluru	14
Rajiv Gandhi International Stadium, Uppal, Hyderabad	13
Maharashtra Cricket Association Stadium, Pune	13
Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket Stadium	13
SuperSport Park	12
Sardar Patel Stadium, Motera	12
Punjab Cricket Association IS Bindra Stadium, Mohali	11
Sawai Mansingh Stadium, Jaipur	10
Saurashtra Cricket Association Stadium	10
Brabourne Stadium	10
Punjab Cricket Association IS Bindra Stadium	10
Himachal Pradesh Cricket Association Stadium	9
MA Chidambaram Stadium	9
Holkar Cricket Stadium	9
New Wanderers Stadium	8
Zayed Cricket Stadium, Abu Dhabi	8
JSCA International Stadium Complex	7
Barabati Stadium	7
St George's Park	7
Newlands	7
Shaheed Veer Narayan Singh International Stadium	6
Nehru Stadium	5
Punjab Cricket Association IS Bindra Stadium, Mohali, Chandigarh	5
Maharaja Yadavindra Singh International Cricket Stadium, Mullanpur	5
Green Park	4
Himachal Pradesh Cricket Association Stadium, Dharamsala	4
Vidarbha Cricket Association Stadium, Jamtha	3
De Beers Diamond Oval	3
Buffalo Park	3
Barsapara Cricket Stadium, Guwahati	3
OUTsurance Oval	2

```
Dr. Y.S. Rajasekhara Reddy ACA-VDCA Cricket Stadium, Visakhapatnam       2
Name: venue, dtype: int64
```

In [24]:
```python
top5_venues = venue.head(5)

print(top5_venues)
```

```
Eden Gardens                                77
Wankhede Stadium                            73
M Chinnaswamy Stadium                       65
Feroz Shah Kotla                            60
Rajiv Gandhi International Stadium, Uppal    49
Name: venue, dtype: int64
```

In [25]:
```python
plt.figure(figsize=(10,5))
top5_venues.plot(kind='bar', color='purple')
plt.title("Top 5 IPL Venues")
plt.xlabel("Venue")
plt.ylabel("Matches")
plt.xticks(rotation=45)
plt.show()
```



In [26]:
```python
print("Largest Margin:", df['result_margin'].max())
```

```
Largest Margin: 146.0
```

In [27]:
```
largest = df[df['result_margin'] == df['result_margin'].max()]

largest
```

Out[27]:

	id	season	city	date	match_type	player_of_match	venue	team1	team2	toss_
620	1082635	2017	Delhi	2017-05-06	League	LMP Simmons	Feroz Shah Kotla	Delhi Daredevils	Mumbai Indians	Dar

In [28]:
```
top10_margin = df.sort_values(by='result_margin', ascending=False).head(10)

top10_margin[['season','team1','team2','winner','result_margin']]
```

Out[28]:

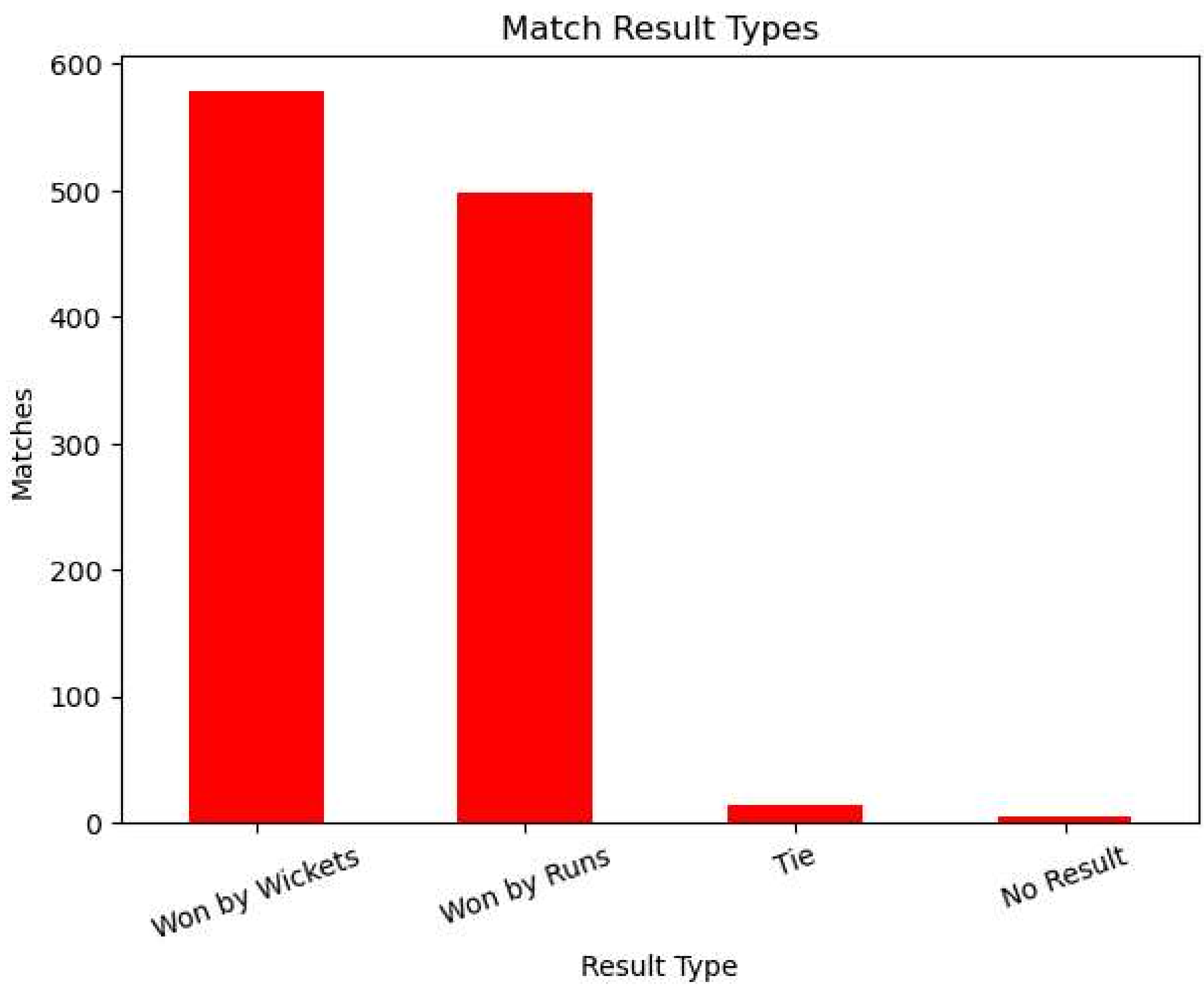
	season	team1	team2	winner	result_margin
620	2017	Delhi Daredevils	Mumbai Indians	Mumbai Indians	146.0
560	2016	Royal Challengers Bangalore	Gujarat Lions	Royal Challengers Bangalore	144.0
0	2007/08	Royal Challengers Bangalore	Kolkata Knight Riders	Kolkata Knight Riders	140.0
496	2015	Royal Challengers Bangalore	Kings XI Punjab	Royal Challengers Bangalore	138.0
352	2013	Royal Challengers Bangalore	Pune Warriors	Royal Challengers Bangalore	130.0
706	2019	Sunrisers Hyderabad	Royal Challengers Bangalore	Sunrisers Hyderabad	118.0
1009	2023	Royal Challengers Bangalore	Rajasthan Royals	Royal Challengers Bangalore	112.0
236	2011	Kings XI Punjab	Royal Challengers Bangalore	Kings XI Punjab	111.0
1039	2024	Kolkata Knight Riders	Delhi Capitals	Kolkata Knight Riders	106.0
55	2007/08	Delhi Daredevils	Rajasthan Royals	Rajasthan Royals	105.0

In [29]:
```
def result_type(row):
    if row['result'] == 'runs':
        return 'Won by Runs'
    elif row['result'] == 'wickets':
        return 'Won by Wickets'
    elif row['result'] == 'tie':
        return 'Tie'
    else:
        return 'No Result'

df['win_type'] = df.apply(result_type, axis=1)
```

In [31]:
```
result_counts = df['win_type'].value_counts()

print(result_counts)
```

```
Won by Wickets    578
Won by Runs       498
Tie                14
No Result           5
Name: win_type, dtype: int64
```

In [32]:
```
plt.figure(figsize=(7,5))
result_counts.plot(kind='bar', color='red')
plt.title("Match Result Types")
plt.xlabel("Result Type")
plt.ylabel("Matches")
plt.xticks(rotation=20)
plt.show()
```



In [33]:
```
df['date'] = pd.to_datetime(df['date'])

print(df['date'].dtype)
```

```
datetime64[ns]
```

In [34]:
```python
df['year'] = df['date'].dt.year

df[['date','year']].head()
```

Out[34]:

	date	year
0	2008-04-18	2008
1	2008-04-19	2008
2	2008-04-19	2008
3	2008-04-20	2008
4	2008-04-20	2008

In [35]:
```python
def win_category(result):
    if result == 'runs':
        return 'Won by Runs'
    elif result == 'wickets':
        return 'Won by Wickets'
    elif result == 'tie':
        return 'Tie'
    else:
        return 'No Result'

df['win_type'] = df['result'].apply(win_category)

df[['result','win_type']].head()
```

Out[35]:

	result	win_type
0	runs	Won by Runs
1	runs	Won by Runs
2	wickets	Won by Wickets
3	wickets	Won by Wickets
4	wickets	Won by Wickets

In [ ]:

In [ ]: